

TYLER POLLARD

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PROFESSIONAL SUMMARY

Statistician with 7+ years of experience in predictive modeling, advanced statistical analysis, and delivering actionable insights. Passionate about sports analytics, with personal projects modeling NFL data to predict scores, spreads, and outcomes using R and Stan. Skilled in developing dynamic R Shiny applications and scalable workflows, combining technical excellence with a commitment to innovation and data-driven decision-making.

SKILLS

Programming & Tools: R, R Shiny, ggplot2, plotly, Stan, JAGS, SQL, JMP, SAS, GitHub, GitLab, Tableau

Statistical Expertise: Predictive Modeling, Regression Modeling, Bayesian Statistics, Multilevel Modeling, Multivariate Modeling, Time Series Analysis, Machine Learning, Design of Experiments (DOE), Data Visualization

Security Clearance: Active Top Secret / Sensitive Compartmented Information

WORK EXPERIENCE

Operations Research Analyst | April 2024 - Current *U.S. Special Operations Command* | Tampa, FL

- Developed Bayesian multilevel ordinal regression models to analyze Likert survey data, enabling iterative updates to improve model accuracy. Delivered actionable insights that informed strategic decision-making and enhanced the evaluation of behavior change.
- Designed and implemented a stance detection framework combining large language models with Bayesian hierarchical modeling. Scaled analysis from 43 text items to tens of thousands, achieving a 32% improvement in prediction accuracy while quantifying uncertainty for more informed decision-making.
- Conducted survival analysis on social media time-to-event data using semi-parametric joint models with time-varying predictors. Identified critical time thresholds that shaped strategic adjustments in planning and execution.
- Programmed and deployed R Shiny applications on the Posit Connect cloud platform, enabling over 50 assessment team members to perform standardized statistical analyses. Improved workflows for data visualization and reporting.
- Promoted to lead a 6-person assessments team, overseeing the development of standardized evaluation frameworks and innovative solutions to assess Military Information Support Operations across 11 combatant commands.

Mathematical Statistician | September 2018 - April 2024 *U.S. Army Evaluation Center* | Aberdeen Proving Ground, MD

- Collaborated with 24 multi-disciplinary teams to develop statistical models, applying DOE, simulations, and advanced techniques. Delivered test designs that addressed critical evaluation metrics and reduced

resource requirements by up to 67%.

- Designed and automated modular workflows for analyzing multilevel probability response variables. Used R and JMP scripts to integrate Monte Carlo simulations, Beta-Binomial regression, and kernel density estimation, enabling scalable and iterative analyses.
- Performed extensive analyses on continuous, discrete, and survey datasets using regression, mixed models, and Bayesian methods. Delivered actionable reports and visualizations that supported data-driven decision-making by senior leadership.
- Applied advanced statistical techniques to ensure the success of a top-priority operational test and transform test infrastructure, earning the “Department of the Army Civilian Service Commendation Medal.”
- Enhanced probabilistic modeling tools, improving the accuracy and efficiency of test design processes. Recognized as the “AEC Employee of the Quarter” for these contributions.

EDUCATION

Master of Statistics | July 2024 *North Carolina State University* | *Raleigh, NC* | *GPA: 4.00*

B.S. in Mechanical Engineering | May 2018 *Clemson University* | *Clemson, SC* | *GPA: 3.53*